Linux System Health Check & Automation

SYSTEM ADMINISTRATION & INFRASTRUCTURE PORTFOLIO

Analyst Name	Analysis Tool	Dataset Log	Analyst Date
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1. Executive Summary

Proyek ini mendokumentasikan pembuatan skrip pemantauan kesehatan sistem otomatis (*system health check*) berbasis Bash di lingkungan Ubuntu Linux. Solusi ini dirancang untuk mendeteksi metrik vital server, mencatat performa secara periodik, serta memastikan kelangsungan layanan kritis tanpa intervensi manual.

Key Monitoring Features

- **Disk Usage:** Memantau penggunaan partisi kapasitas penyimpanan utama dan boot.
- **Memory & Swap:** Melacak pemakaian RAM dan Swap memory secara real-time.
- **CPU Load Average:** Menghitung beban pemrosesan sistem (1, 5, 15 menit).
- **SSH Service Check:** Verifikasi status aktif layanan OpenSSH Daemon.

Technical Specifications

- **OS Platform:** Ubuntu 24.04 LTS (VirtualBox VM).
- **Shell Environment:** GNU Bash.
- **Task Automation:** Crontab (Daemon Service).
- **User Privileges:** Non-root (`infra`) for security.

2. Automation & Execution Setup

Step/Task	Command	Description
Execution Permission	chmod +x system_health_check.sh	Memberikan izin eksekusi skrip Bash
Manual Execution	./system_health_check.sh	Menjalankan pengujian langsung di terminal
Cron Job Scheduling	0 0 * * * /home/infra/system_health_check.sh	Menjadwalkan eksekusi otomatis setiap hari pukul 00:00 AM

3. Sample Diagnostic Output

```
infra@infra:~$ chmod +x system_health_check.sh
infra@infra:~$ ./system_health_check.sh
=== SYSTEM HEALTH CHECK ===
Date: Fri Oct 30 10:16:10 AM UTC 2026
[*] Disk Usage:
/dev/mapper/ubuntu--vg-ubuntu--lv   12G  7.3G  3.4G  69% /
/dev/sda2                          2.0G  201M  1.6G  11% /boot
[*] Memory Usage:
               total        used        free      shared  buff/cache   available
Mem:            3915         447        3312           1         373        3468
Swap:           2047           0        2047
[*] CPU Load:
 10:16:10 up 8 min,  1 user,  load average: 0.10, 0.08, 0.06
[*] SSH Service Status:
active
===============
infra@infra:~$ crontab -e
no crontab for infra - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 0 0 * * * /home/infra/system_health_check.sh
Choose 1-4 [1]: _
```

```
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command

0 0 * * * /home/infra/system_health_check.sh









crontab: installing new crontab
infra@infra:~$
```

4. Troubleshooting & System Best Practices

Operational Diagnostic

- **Absolute Pathing:** Menggunakan (`/home/infra/system_health_check.sh`) *absolute path* pada Crontab untuk mencegah masalah eksekusi direktori kerja..
- **Log File Management:** Menyiapkan keluaran hasil pengecekan secara beruntun (*append*) ke file log terpusat.

Security & Hardening

- **Principle of Least Privilege:** Skrip dijalankan di bawah akun pengguna terbatas (`infra`) tanpa bergantung pada akses `root`.
- **Service Availability:** Pemantauan status SSH memastikan akses manajemen jarak jauh server tetap aman dan tersedia.

5. Future Enhancement Roadmap

- **Alerting Integration:** Mengintegrasikan skrip dengan Telegram Bot API atau Webhook Slack untuk mengirim notifikasi instan ketika kapasitas disk > 80% atau memori kritis.
- **Log Rotation (logrotate):** Mengonfigurasi mekanisme rotasi log otomatis guna mengompresi dan menghapus log historis untuk mencegah kepenuhan media penyimpanan.
- **Multi-Service Monitoring:** Memperluas deteksi ke layanan pendukung lainnya seperti Web Server (Nginx/Apache) dan Database (MySQL/PostgreSQL).